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ANALOGY

PART 3

Number Analogy | Concepts, Patterns, Shortcuts & Examiner's Tricks

Competitive Exams: Punjab Police, Punjab Police SI, PSSSB, Punjab PCS, SSC, Banking, Railways, CDS, CAPF

Introduction

What is Number Analogy?

A **Number Analogy** compares two numbers using a mathematical relationship.

Your task is to identify the relationship in the first pair and apply the **same rule** to the second pair.

Golden Rule for Number Analogy

Always check the operations in this order:

1. Addition (+)
2. Subtraction (−)
3. Multiplication (×)
4. Division (÷)
5. Square (x^2)
6. Cube (x^3)
7. Square Root ($\sqrt{}$)
8. Prime Numbers
9. Odd–Even Pattern
10. Difference Pattern

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Tip: Never guess the answer. First identify the exact mathematical relationship.

Type 1 – Addition

Simple Explanation

The second number is obtained by adding a fixed value.

Example

$$15 : 20 :: 32 : ?$$

Relation:

$$20 = 15 + 5$$

Answer:

$$32 + 5 = 37$$

Exam Tip

Check addition first because it is the simplest pattern.

Type 2 – Subtraction**Simple Explanation**

The second number is obtained by subtracting a fixed value.

Example

$$25 : 18 :: 40 : ?$$

Relation:

$$18 = 25 - 7$$

Answer:

$$40 - 7 = 33$$

Type 3 – Multiplication**Simple Explanation**

Multiply the first number by a fixed number.

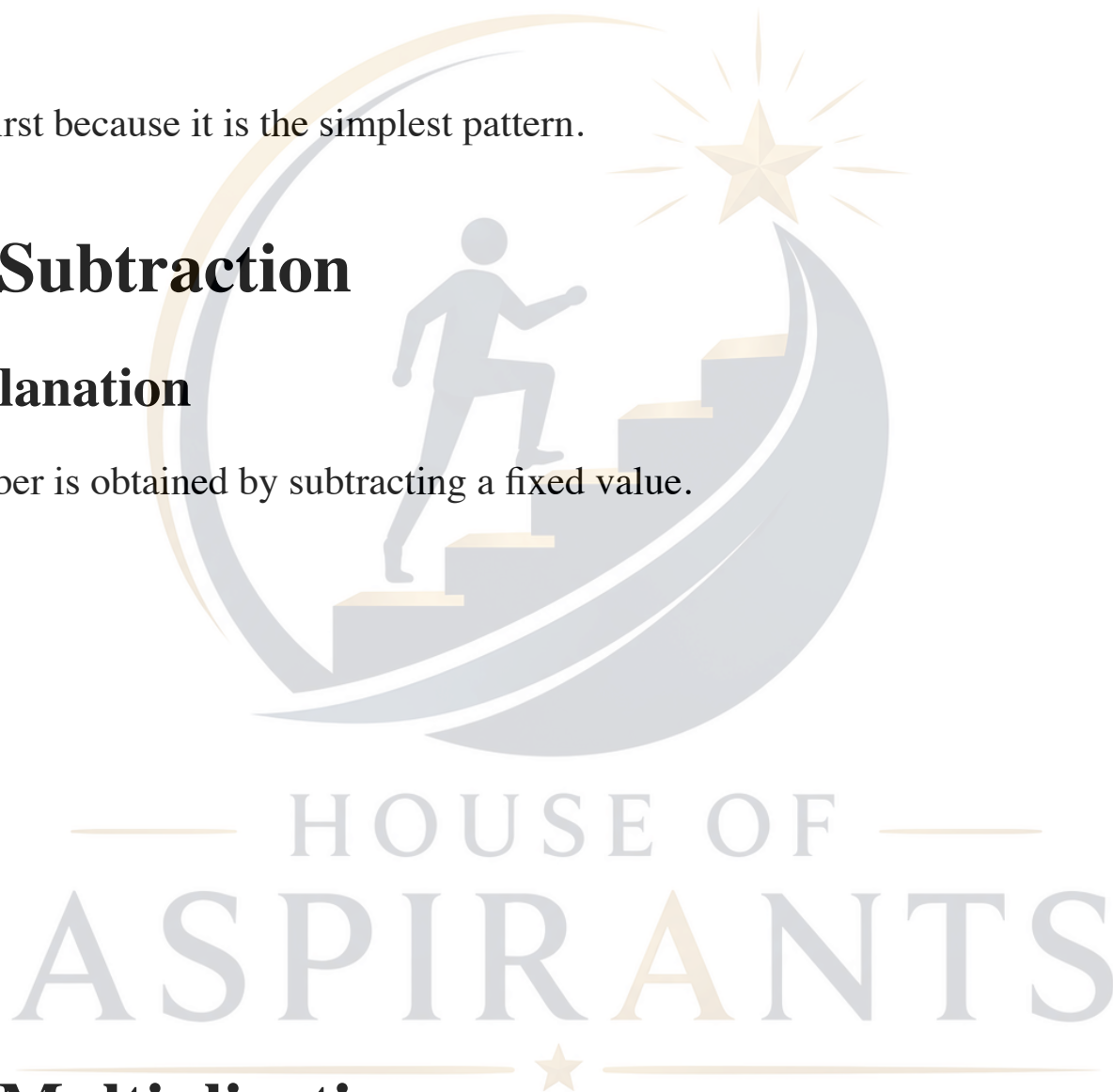
Example

$$8 : 24 :: 12 : ?$$

Relation:

$$24 = 8 \times 3$$

Answer:



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$$12 \times 3 = 36$$

Type 4 – Division

Simple Explanation

The second number is obtained by dividing the first number.

Example

$$36 : 12 :: 45 : ?$$

Relation:

$$36 \div 3 = 12$$

Answer:

$$45 \div 3 = 15$$

Type 5 – Square

Simple Explanation

Square means multiplying the number by itself.

Formula

$$n^2 = n \times n$$

Important Squares

Number	Square
2	4
3	9
4	16
5	25
6	36
7	49
8	64
9	81
10	100
11	121
12	144

13	169
14	196
15	225

Example

9 : 81 :: 12 : ?

Answer:

144

Type 6 – Cube

Simple Explanation

Cube means multiplying the number three times.

Formula

$$n^3 = n \times n \times n$$

Important Cubes

Number	Cube
2	8
3	27
4	64
5	125
6	216
7	343
8	512
9	729
10	1000

Example

4 : 64 :: 5 : ?

Answer:

125

Type 7 – Square Root

Simple Explanation

Square root is the number which, when multiplied by itself, gives the original number.

Important Square Roots

Number	Square Root
25	5
36	6
49	7
64	8
81	9
100	10
121	11
144	12
169	13
196	14

Example

$49 : 7 :: 81 : ?$

Answer:

9



Type 8 – Prime Number Pattern

Simple Explanation

Prime numbers have only **two factors**.

Important Prime Numbers

2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31

Example
 $11 : 13 :: 17 : ?$

Answer:

19**Type 9 – Odd & Even Pattern****Simple Explanation**

Questions are based on odd or even number sequences.

Odd Numbers

1, 3, 5, 7, 9, 11, 13

Even Numbers

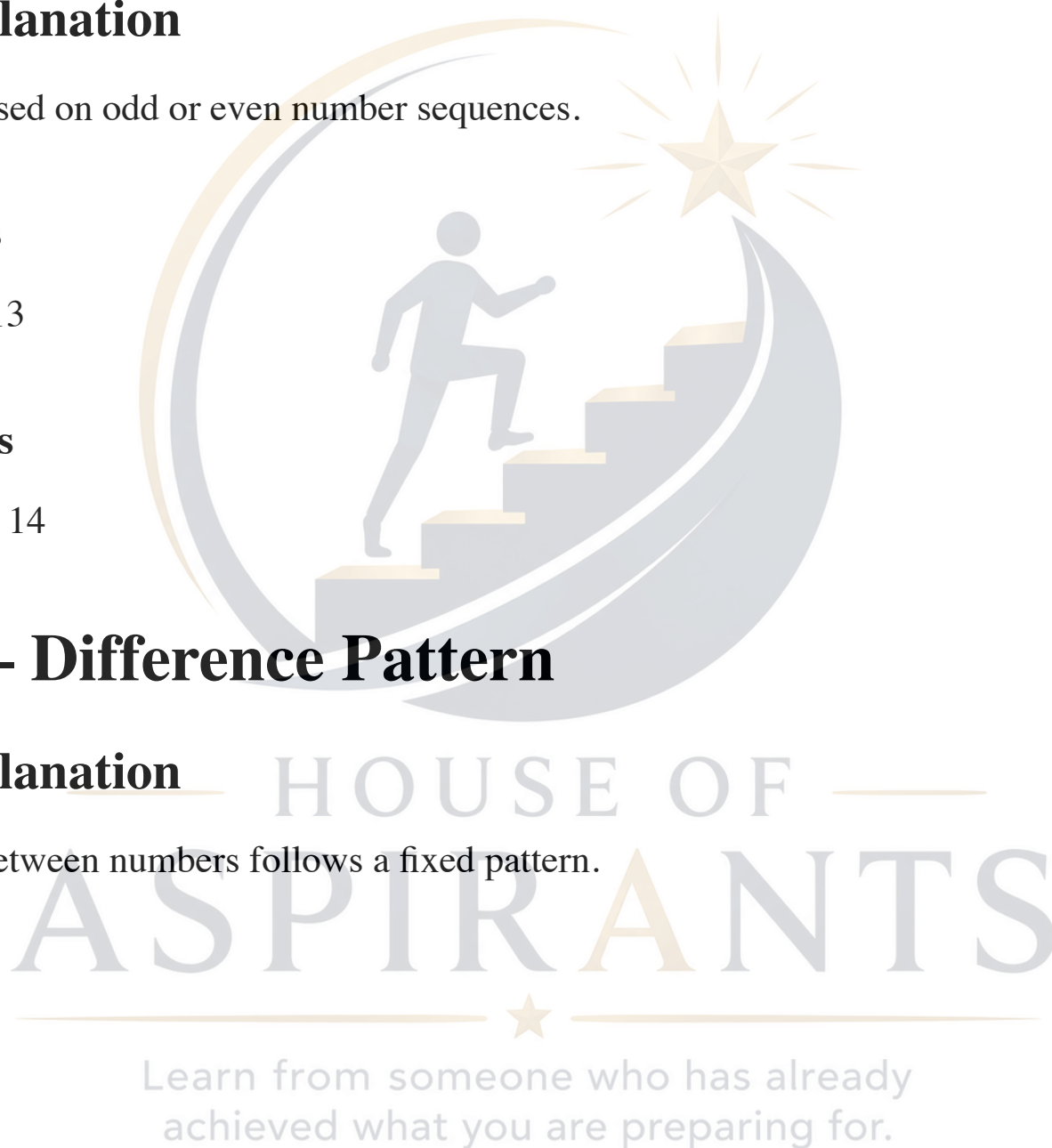
2, 4, 6, 8, 10, 12, 14

Type 10 – Difference Pattern**Simple Explanation**

The difference between numbers follows a fixed pattern.

Example
 $6 : 10 :: 12 : ?$

Difference = +4

Answer = **16****Most Common Number Analogy Patterns**

Pattern	Example
+ Constant	12 → 18
– Constant	25 → 20
× Constant	8 → 24
÷ Constant	36 → 12

Square	9 → 81
Cube	4 → 64
Square Root	49 → 7
Prime Series	11 → 13
Odd-Even	7 → 9
Difference	4

Examiner's Favourite Tricks

Trick 1

Check **Multiplication before Addition** when numbers are large.

Trick 2

If numbers increase rapidly, check:

- Square
- Cube
- Multiplication

Trick 3

If numbers decrease rapidly, check:

- Division
- Square Root

Trick 4

If no operation matches, check:

- Prime Numbers
- Odd-Even
- Difference Pattern

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Common Mistakes

✗ Ignoring multiplication and checking only addition.

✗ Missing square or cube relationships.

✗ Confusing square with cube.

✗ Forgetting square roots.

✗ Applying different rules to different pairs.

Memory Tricks

Squares

" $15^2 = 225$ "

Remember squares from **1–20**.

Cubes

" $5^3 = 125$ "

Remember cubes from **1–10**.

Prime Numbers

"**2** is the only even prime."

Order Rule

$+$ $\rightarrow$ $-$ $\rightarrow$ $\times$ $\rightarrow$ $\div$ $\rightarrow$ **Square** $\rightarrow$ **Cube** $\rightarrow \sqrt{\quad}$

Always check in this sequence.

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